Alwan Muhammad Rasyid

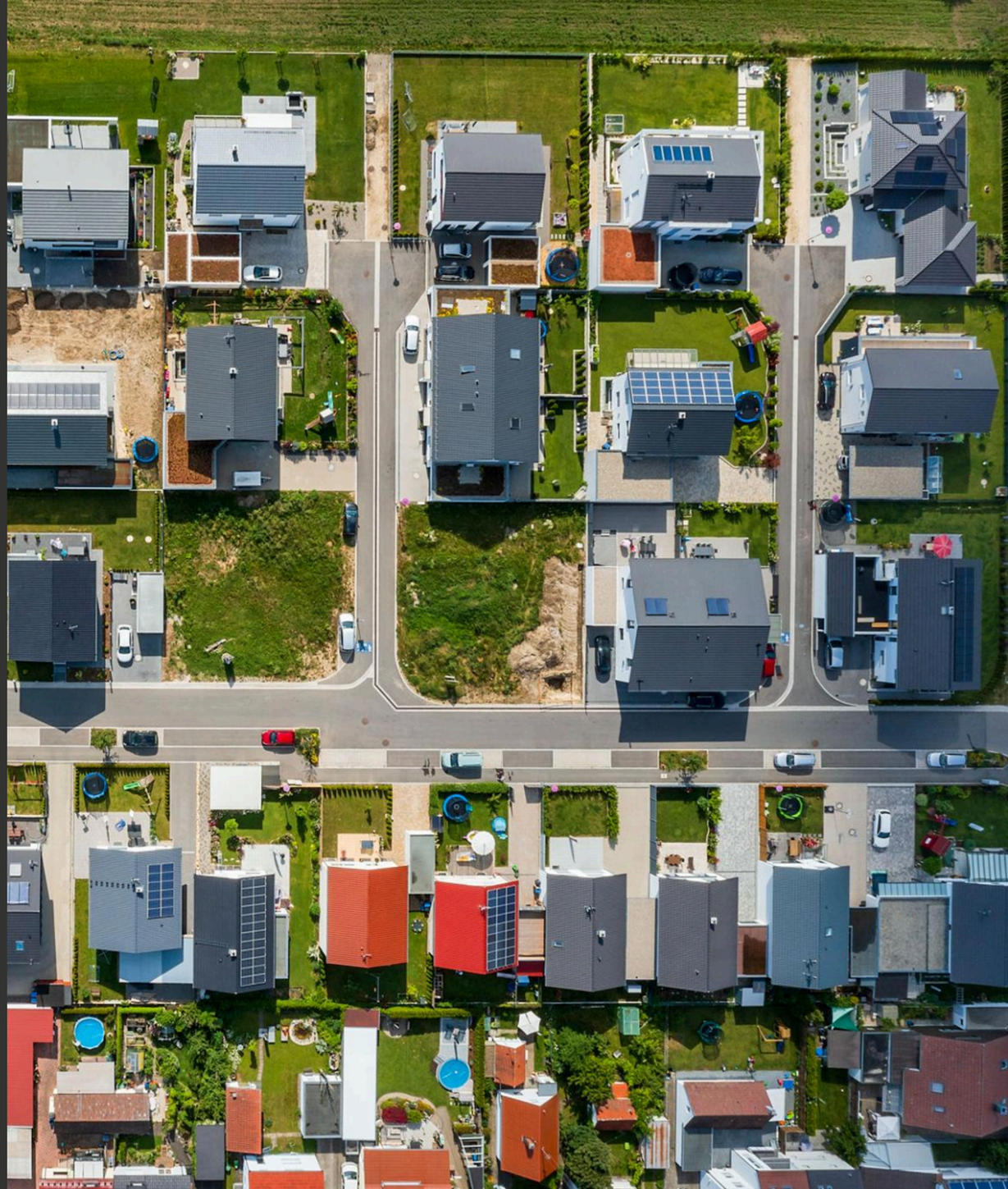
Housing Value Prediction

With Regularized Regression

Evaluating Multicollinearity, Target Treatment, and Model Trade-offs

Machine Learning Portfolio Project

Python · Scikit-learn · Regression Modeling



Career Direction

Building a career toward AI Engineering, with Data Science and Machine Learning as the foundation for developing expertise in predictive modeling, applied AI, and data-driven problem solving.

Education

- Institut Teknologi Sepuluh Nopember (ITS) — B.Eng. Geophysical Engineering

↓

- Dibimbing.id — Data Science & AI/ML Bootcamp

Experience

Experience spanning commercial banking and geophysical engineering, complemented by exposure to data acquisition, technical analysis, business development, and cross-functional collaboration.

- Management Development Associate Intern · **Bank Mandiri**
- Geotechnical Engineer Intern · **Virama Karya**
- Field Assistant · **ITS**
- Head of External Relation · **HMTG ITS**



About Me

Alwan Muhammad Rasyid

Geophysical Engineering graduate with experience in technical analysis, fieldwork, and commercial banking. Currently expanding technical expertise through a Data Science & AI/ML Bootcamp, with interests in data analytics, predictive modeling, and data-driven decision making.

Skills

Data	Python · SQL · Pandas · Data Visualization · Exploratory Data Analysis
Machine Learning	Regression · Classification · Model Evaluation · Feature Engineering
Tools	Scikit-learn · Jupyter Notebook · Git · DBeaver

Estimating **housing value** can support data-driven **property valuation .**

Business Context

Property valuation can involve many observable characteristics, such as property size, accessibility, environmental conditions, and neighborhood attributes.

A regression model can help estimate housing value from these characteristics and provide an initial, data-driven reference for valuation analysis.

Potential Use

Provide a consistent, data-driven estimate that can support preliminary valuation and further analysis

Potential users

- Property valuation teams
- Real-estate analysts
- Housing market researchers

1

Property & Neighborhood Characteristics

2

Regression Model

3

Estimated Housing Value

4

Valuation Decision Support

How can **housing value** be **predicted reliably** from property and neighborhood characteristics?

The project evaluates whether feature selection, regularization, and target treatment can improve predictive performance while considering their modeling trade-offs.

Problem Statement

Build a regression model to **predict medv** (median value of owner-occupied homes) using 13 property and neighborhood characteristics.

Model performance is evaluated using **RMSE**, **MAE**, and **MAPE** on unseen data.

Project objectives

1

Establish a baseline

Compare **Ridge** and **LASSO** regression as regularized baseline models.

2

Test feature selection

Evaluate whether **removing a highly correlated feature** improves predictive performance.

3

Investigate target treatment

Assess the impact of observations at the apparent **medv = 50** upper boundary.

4

Compare modeling approaches

Determine whether **regularization** provides a meaningful advantage over **Multiple Linear Regression.**

The dataset provides **13 measurable characteristics** to predict housing value

Dataset Overview

506
observations

13
predictor

medv
Median value of owner-occupied homes
target variable

split
test / validation / train

64% / 16% / 20%

Data Quality

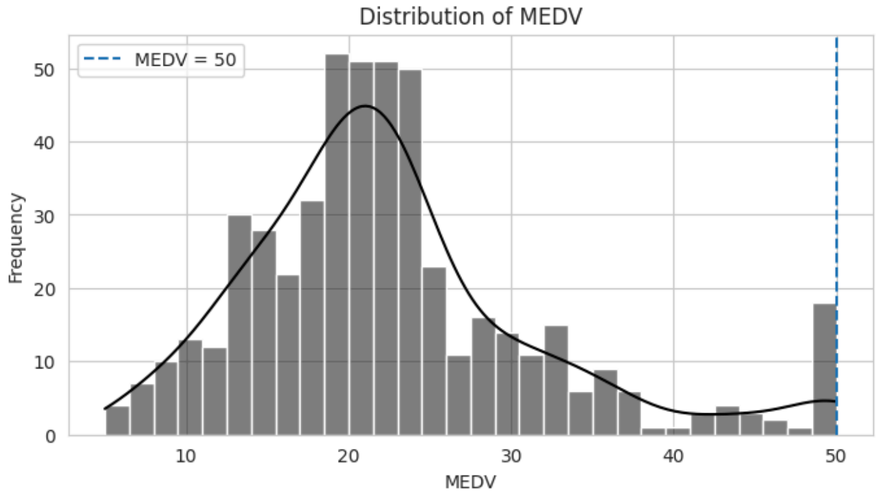
- Missing values: **0**
- Duplicates: **0**
- Major data-quality issues: **None identified**

Feature Categories

- Property Characteristics
  rm · age
- Accessibility
  rad · tax
- Neighborhood / Socioeconomic
  crim · zn · indus · chas · dis
  ptratio · black · lstat
- Environmental
  nox

Two patterns in the data shaped the modelling way

Target Distribution



Finding 01

medv shows an apparent upper boundary around 50.

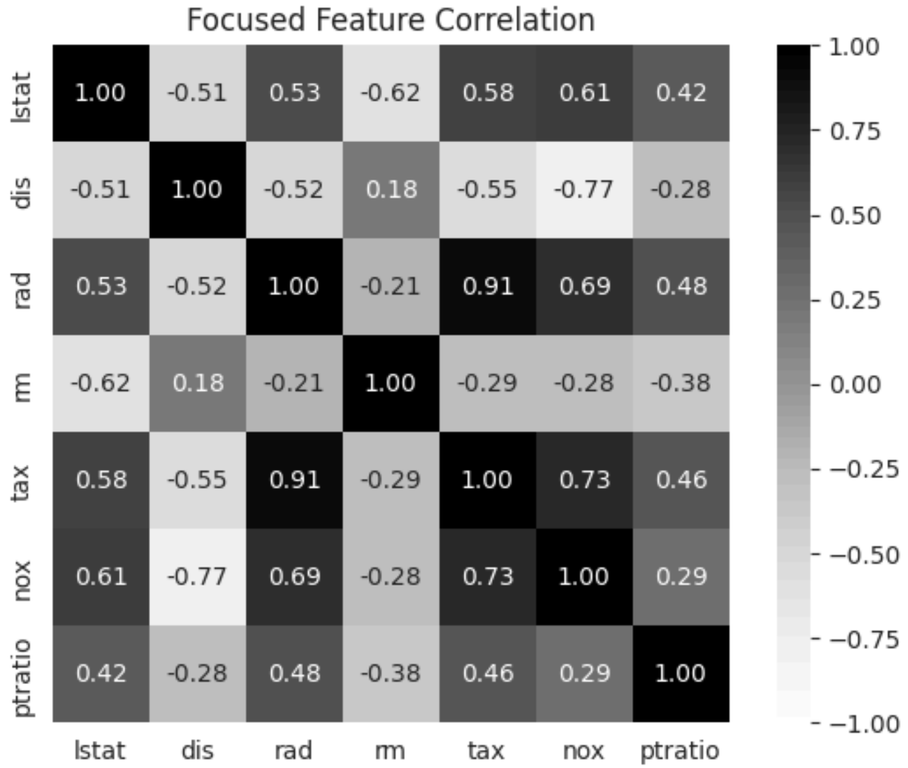
- *This motivated a sensitivity experiment on observations at the upper boundary.*

Finding 02

rad and **tax** show a strong correlation (~0.91), accompanied by relatively high VIFs.

- *This motivated a feature-selection experiment.*

Focused Correlation and Multicollinearity



Multicollinearity:

	Feature	VIF score
1	rad	**7.808406**
2	tax	**8.842757**

Two **modeling challenges** to targeted experiments

Rather than assuming the solution, each modeling concern was tested against validation performance.

Challenge 1: Multicollinearity

Evidence

rad and **tax** show a strong correlation (~0.91), with relatively high VIFs.

Modeling question

Does removing **rad** improve predictive performance?

Experiment

Full features
vs.
Remove **rad**

Challenge 2: Apparent Target Censoring

Evidence

Dataset documentation indicates that **medv** appears censored at 50.00.

Modeling question

Does treating observations at the upper boundary improve predictive performance?

Experiment

Original target
vs.
Exclude **medv ≥ 50**

StandardScaler → Model Training → Validation RMSE → Select Approach → Test Evaluation

Ridge and **LASSO** delivered nearly identical baseline performance

1 Modeling setup

Models

- **Ridge** Regression
- **LASSO** Regression

Preprocessing

- StandardScaler
- Full 13 features

Hyperparameter tuning

- Alpha selected using validation RMSE

Data split

- 64% Train
- 16% Validation
- 20% Test

2 Validation result

	Best α	Validation RMSE
Ridge	1	4.6267
LASSO	0.01	4.6279

3 Test result

	RMSE ↓	MAE ↓	MAPE ↓
Ridge	4.3058	3.3248	17.9628%
LASSO	4.3021	3.3221	17.9608%

Key Insight & Takeaway

- **LASSO** slightly outperformed Ridge, but the difference was negligible in practical terms.
- At baseline, model choice between **Ridge** and **LASSO** had little impact on predictive performance.
- **Model choice alone did not materially change baseline predictive performance**

Ridge and **LASSO** established a similar baseline, giving us a reference point for the experiments that followed.

Removing the correlated feature **did not improve** predictive performance

Experiment Setup

- Question

  Does removing **rad** reduce multicollinearity while improving prediction?

- Hypothesis

  Removing **rad**, which is highly correlated with **tax**, may reduce multicollinearity and improve model performance.

- Experiment

  Full 13 features vs. Remove **rad**

- Evaluation

  Validation **RMSE**

Validation Result

	Best Model	RMSE
Full features	Ridge, α=1	4.6267
Remove **rad**	LASSO, α=0.01	4.7006

Decision

Keep the full feature set.

- High correlation between predictors does not automatically justify feature removal when the primary objective is predictive performance.

Treating the apparent target boundary **improved** predictive performance

Observations at the upper boundary of **medv** were excluded to assess their impact on model performance.

1 What was the issue?

Dataset documentation indicates that **medv** appears to be **censored** at **50.00**.

2 Treatment

Original data → Exclude → Treated data

506 observations | medv ≥ 50 | 490 observations

16 observations removed · 3.16% of dataset

3 Validation Result

Improvement: ~0.9065 RMSE

	Best Model	RMSE
Original Target	Ridge, α=1	4.6267
Treated target	Ridge, α=100	3.7202

4 Test Result

	RMSE	MAE	MAPE
Baseline Ridge	4.3058	3.3248	17.9628%
Treated Ridge	4.0052	3.0258	15.8895

Key Takeaway

Target treatment had a larger impact on predictive performance than the choice between Ridge and LASSO.

Better metrics came with **narrower** target coverage

The improvement after target treatment must be interpreted together with what the model is no longer trained to predict.

Modeling question

Does treating observations at the upper boundary improve predictive performance?

Experiment

Original target

vs.

Exclude **medv ≥ 50**

Performance Improved

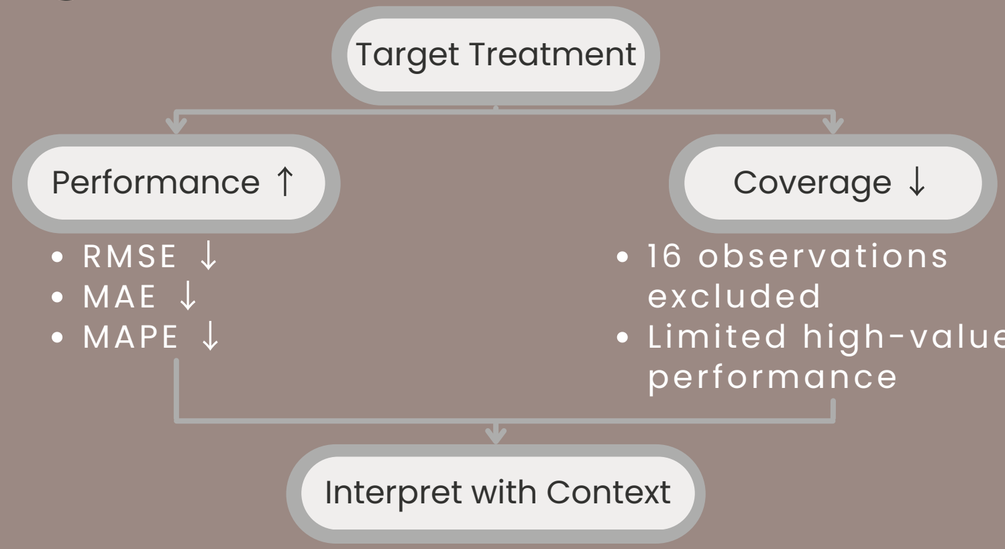
	Original Target	Treated Target
Observations	506	490
Test RMSE	4.3058	4.0052
Test MAE	3.3248	3.0258
Test MAPE	17.9628%	15.8895%

But what changed?

medv < 50 + **medv** ≥ 50 → **medv** < 50

Better performance within a narrower target range

Modeling trade-off



Key Takeaway

Metric improvement does not automatically mean broader predictive capability. The treatment improves performance for the retained target range, but reduces the model's coverage of high-value properties.

Regularization **did not** materially improve predictive performance

After target treatment, LASSO and MLR produced nearly identical predictions, while Ridge showed a different metric trade-off.

Experimental setup

- **Target**: **medv < 50**
- **Features**: all 13 predictors
- **Models**: Multiple Linear Regression, Ridge, LASSO
- **Evaluation**: unseen test set
- **Hyperparameter**: Best proven by baseline and previous experiment

Test Performance

	RMSE	MAE	MAPE
MLR	3.9042	3.1158	17.2670%
Ridge	4.0052	3.0258	15.8895%
LASSO	3.9041	3.1156	17.2661%

Key Takeaway

- Regularization was not automatically beneficial: LASSO performed almost identically to MLR, while Ridge offered lower MAE and MAPE at the cost of higher RMSE.

The best modeling decision **is not simply** the model with the lowest RMSE.

1 What did the experiments show?

1. Keep the full feature set

Removing **rad** despite its high correlation with **tax** did not improve validation performance.

Decision

retain all 13 predictors.

2. Target treatment improved performance

Excluding observations at the apparent **medv = 50** boundary improved test performance.

Decision

treatment can be useful when the modeling objective is limited to the uncensored target range.

3. Regularization is not automatically better

LASSO and **MLR** produced virtually identical results, while **Ridge** offered a different trade-off across evaluation metrics.

Decision

model selection should be based on the metric and business objective, not simply the use of regularization.

2 Final recommendation

For **predicting** housing values within the retained target range, the final model should be selected based on the **primary error metric and intended use**.

3 Business implication

The model can serve as an **initial data-driven reference** for property valuation analysis, **helping** analysts establish a consistent estimate before deeper valuation assessment.

4 Final takeaway

The key lesson is that better modeling comes from validating assumptions and understanding trade-offs, not simply choosing the most complex or most regularized model.

Let’s Connect

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